CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATION SYSTEM USING AGENTFORCE

Salesforce Implementation Project Documentation

1. INTRODUCTION

1.1 Project Overview

The Customer Support Ticket Priority Prediction and Automation System is a Salesforce-based CRM solution developed to streamline the handling of customer support tickets. The system centralizes support ticket data using a custom object (Support Ticket Intelligence), automates the prioritization process based on keyword analysis in ticket descriptions, and utilizes AI-driven Agentforce capabilities to query and summarize data.

This project demonstrates the end-to-end capabilities of Salesforce, including custom object creation, data migration, backend automation via Flows, analytical reporting via Dashboards, and AI integration through Agentforce.

1.2 Purpose

The primary purpose of this system is to:

- Eliminate manual triage of support tickets.
- Automatically categorize tickets into High, Medium, and Low priority based on text analysis.
- Automatically generate follow-up tasks for high-priority issues and assign them to senior agents.
- Provide real-time visibility into ticket statuses and priorities through dynamic dashboards.
- Leverage AI (Agentforce) to allow users to query and summarize support data conversationally.

2. IDEATION & REQUIREMENT ANALYSIS

2.1 Problem Statement

- Manual Triage: Support agents spend too much time manually reading and prioritizing tickets.
- Missed Urgent Issues: Critical issues containing words like "failure" or "urgent" can get lost in the queue, leading to delayed response times.
- Lack of Visibility: Managers lack a centralized view of how tickets are distributed across different priority levels and statuses.

2.2 Solution Requirements

Functional Requirements:

- Create a custom object Support Ticket Intelligence to store ticket data.
- Implement data migration for bulk uploading test ticket data.
- Build an Auto-Launched Flow to analyze descriptions and assign priority.
- Create Reports and Dashboards for data visualization.
- Configure and test an Agentforce AI agent.
- Non-Functional Requirements:
- Ensure data accuracy through strict Picklist restrictions.
- Ensure the system is scalable and easy to maintain.

3. PROJECT DESIGN & ARCHITECTURE

3.1 Data Model

- Custom Object: Support Ticket Intelligence (API Name:Support_Ticket_Intelligence__c)

- Key Fields:
    1. Name (Text): Ticket Number (e.g., TKT-001).
    2. Priority_Level__c (Picklist): High, Medium, Low.
    3. Status__c (Picklist): New, In Progress, Closed.
    4. Issue_Type__c (Picklist): Technical, Billing, General.
    5. Description__c (Long Text Area): Details of the issue.

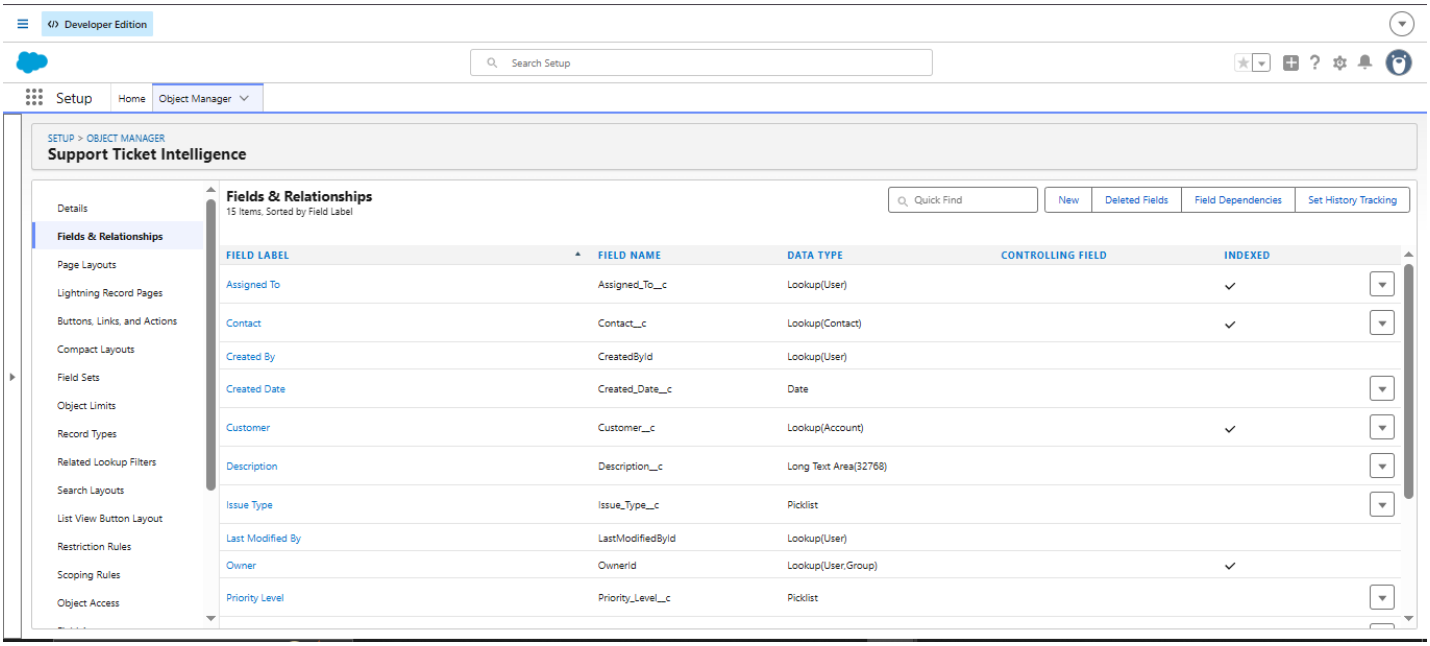


Figure 1.1: Custom Object Data Model for Support Ticket Intelligence

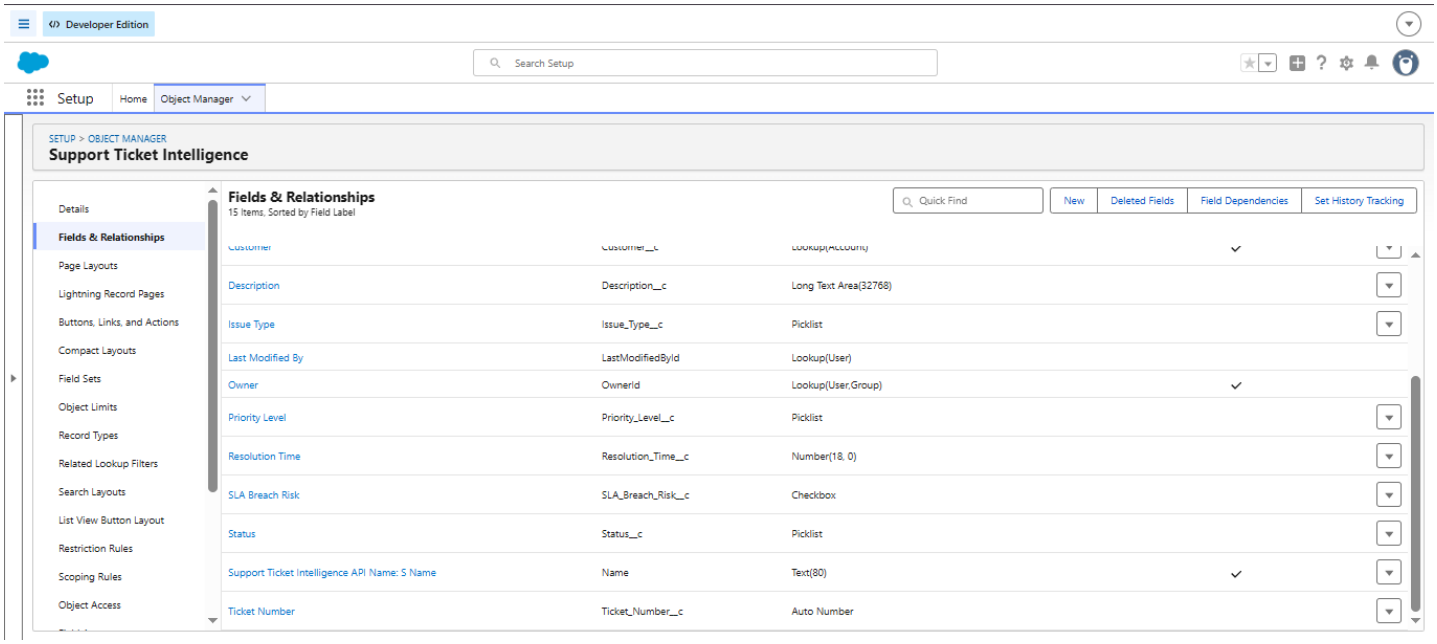


Figure 1.2: Custom Object Data Model for Support Ticket Intelligence

3.2 Business Logic & Automation

- Flow: A custom Auto-Launched Flow (Support_Ticket_Intelligence) that uses Decision elements to search for keywords (urgent, not working, failure, slow, delay) in the Description field. Based on these keywords, it assigns a priority level, creates a Task (for High priority), and outputs a custom action message.

- AI Integration: Agentforce (Default) Agent configured to read the custom object data and provide summaries.

4. PROJECT DEVELOPMENT PHASE (MILESTONES)

4.1 Milestone 1: Data Model & Field Creation

I created the custom object Support Ticket Intelligence and defined the necessary custom fields. To ensure data quality, I configured Priority Level and Status as restricted picklists.

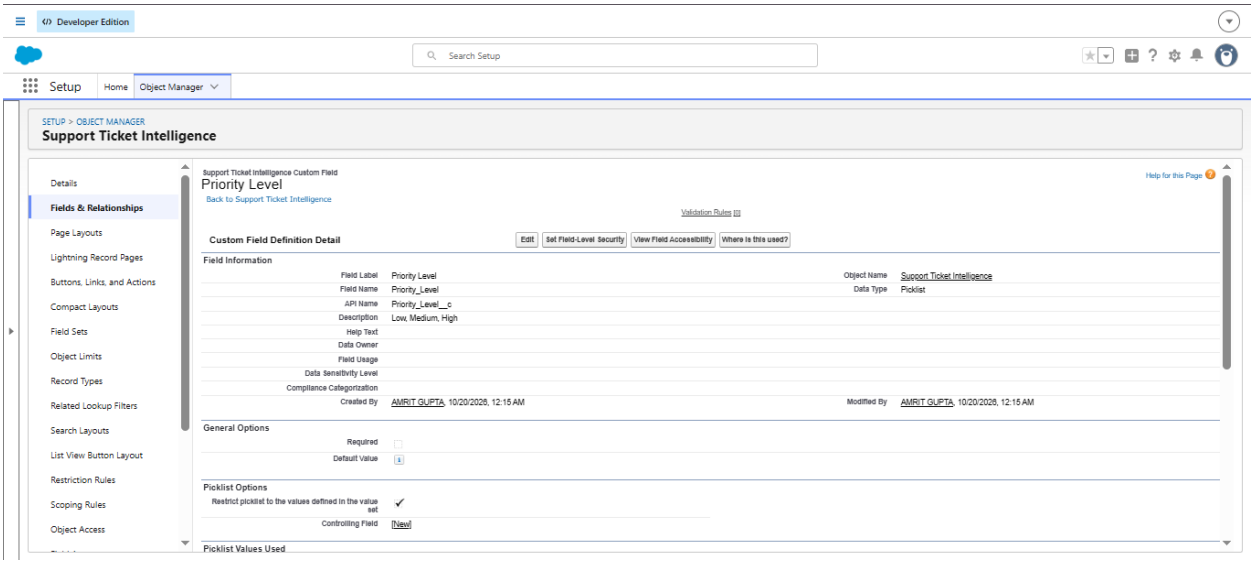


Figure 2.1: Configuration of Restricted Picklist values for Priority Level

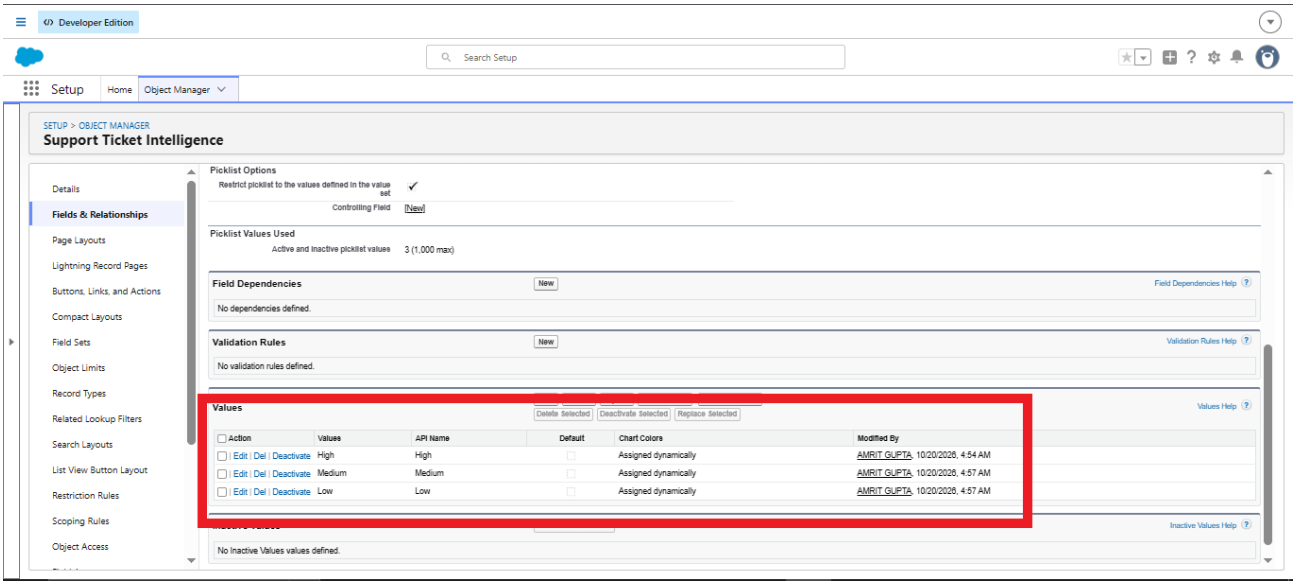


Figure 2.2: Configuration of Restricted Picklist values for Priority Level

4.2 Milestone 2: Data Import (Data Migration)

I prepared a CSV file with 20 sample records containing varied descriptions to test the automation. Using the Data Import Wizard, I attempted to map the fields and import the data into the custom object.

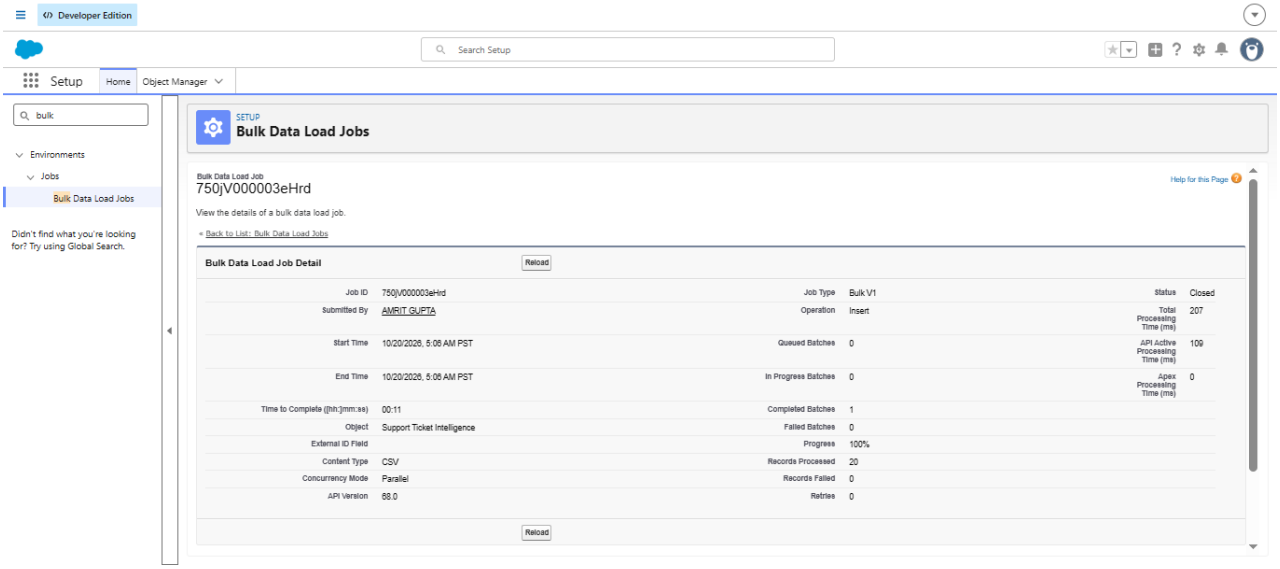


Figure 3: Successful Data Import of 20 records via Data Import Wizard

4.3 Milestone 3: Flow Automation

I built an Auto-Launched Flow with the following logic paths:

- Decision Element: Analyzes the ticket description.
- High Priority Path: If the description contains "urgent", "not working", or "failure", the flow sets the priority to High, creates a Task ("Urgent Ticket Handling"), assigns it to a "Senior Support Agent", and sets a final message.
- Medium Priority Path: If the description contains "issue", "slow", or "delay", the priority is set to Medium.
- Default Path: Any other description is set to Low priority.

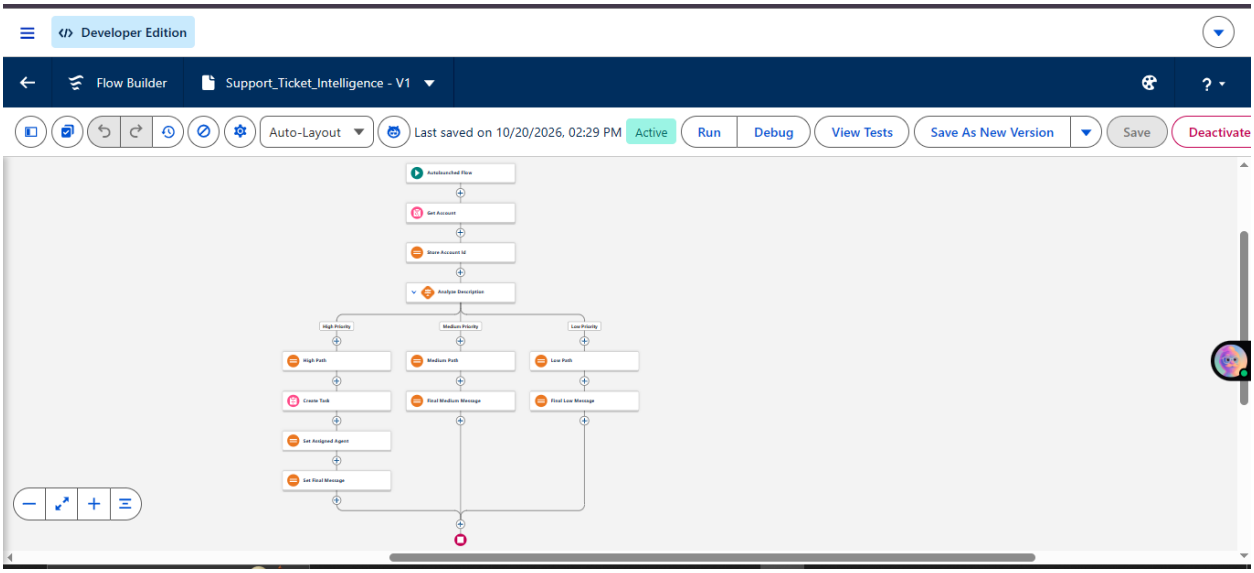


Figure 4: Auto-Launched Flow logic for Ticket Priority Prediction

4.4 Milestone 4: Analytics (Reports & Dashboards)

I created two primary reports:

1. Tickets by Priority Level: Grouped by Priority, displayed as a Donut Chart.
2. Tickets by Status: Grouped by Status, displayed as a Funnel Chart.

These reports were added to a new dashboard named Support Ticket Intelligence Dashboard. Note: In the new Salesforce Lightning UI, widgets are added using the + Add Widget button.

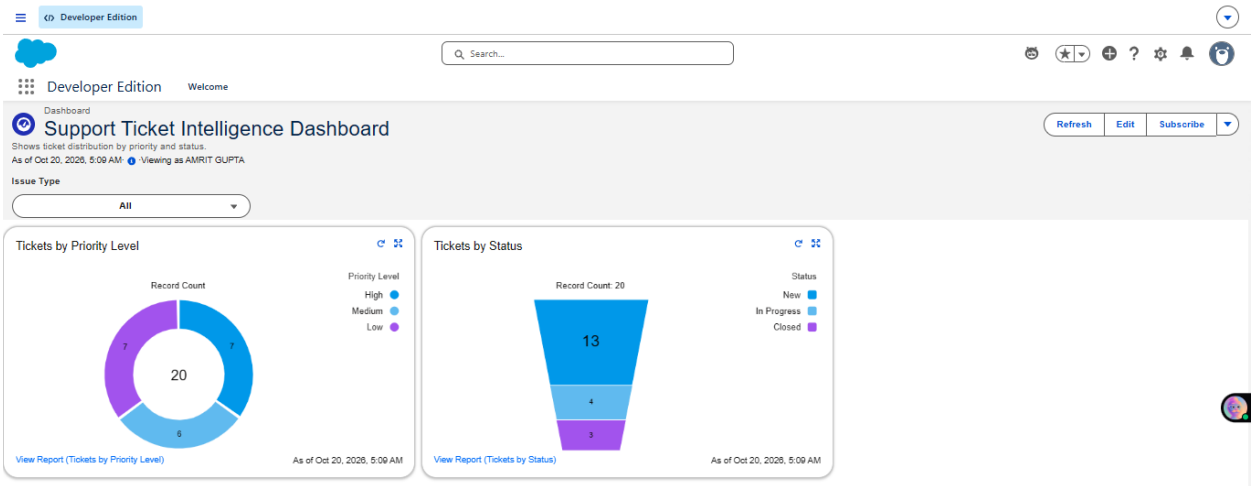


Figure 5: Support Ticket Intelligence Dashboard displaying real-time metrics.

4.5 Milestone 5: Agentforce AI Integration

I activated the Agentforce (Default) agent within the Agent Builder. I used the Conversation Preview to test the AI's ability to query the new object.

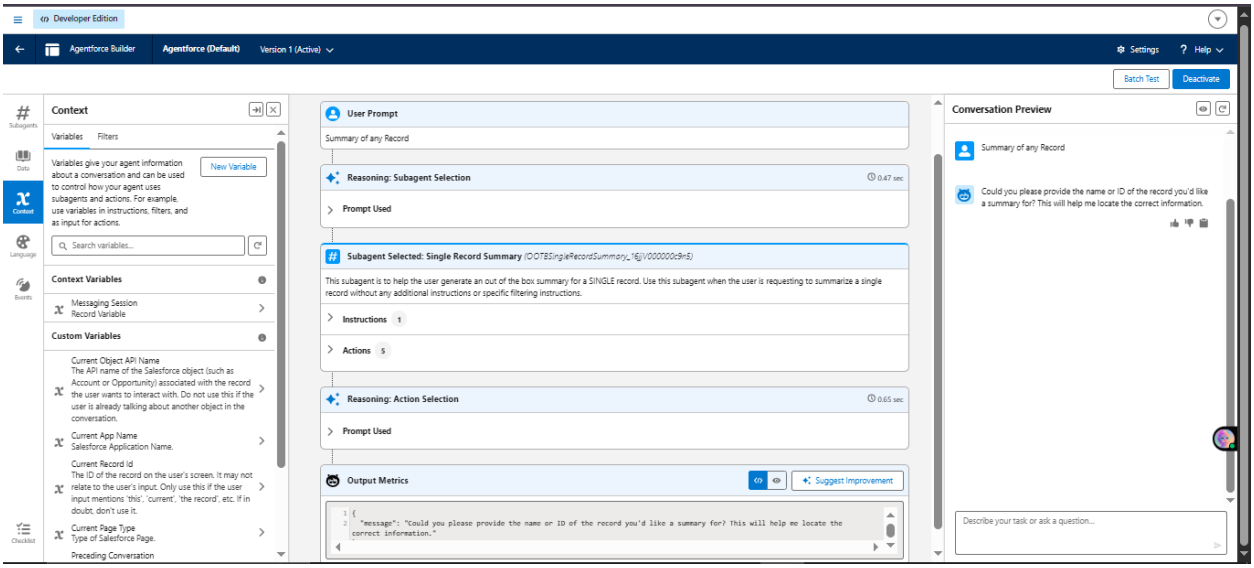


Figure 6: Testing the Agentforce AI agent in the Conversation Preview.

5. TESTING, TROUBLESHOOTING & PROBLEMS FACED

During the implementation, I encountered several technical challenges which I successfully resolved:

Problem 1: Data Import Wizard Failures (Picklist Errors)

- The Issue: My initial attempt to import 20 records resulted in a 100% failure rate. The Bulk Data Load Jobs error log showed: INVALID_OR_NULL_FOR_RESTRICTED_PICKLIST:Priority Level: bad value for restricted picklist field: High.
- The Solution: I realized that when creating the Priority Level field, I had accidentally entered Low, Medium, High as one single string value, rather than three separate picklist values.
- Resolution: I went to Object Manager > Priority Level > Values. I deleted the combined string and added three separate values: High, Medium, and Low. I did the same for the Status field (New, In Progress, Closed).

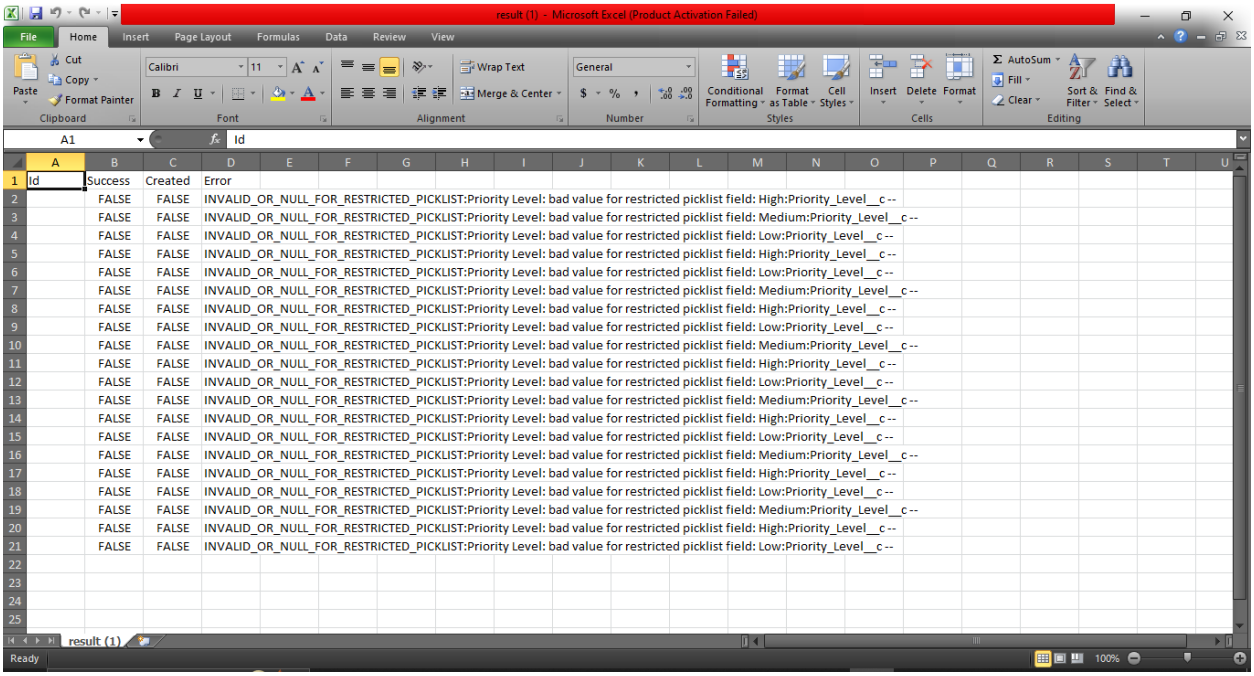


Figure 7.1: Before problem solve import failure log due to restricted picklist errors.

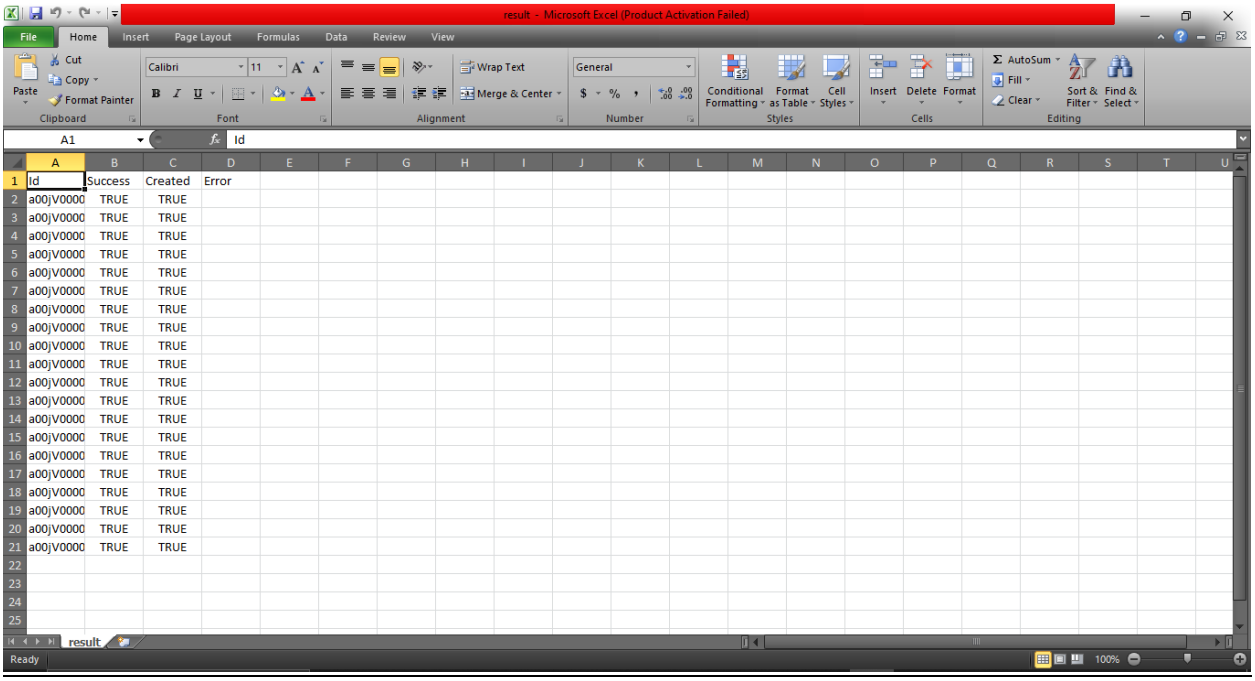


Figure 7.2: After problem solve import successfully.

Problem 2: CSV Field Mapping Issues

- The Issue: During the import, the Customer and Assigned To columns showed as "Unmapped."
- The Solution: I learned that the Data Import Wizard struggles to map Lookup relationships (like Customer) using text names, and mapping the Owner field requires a 15-character Salesforce ID.
- Resolution: I removed the problematic columns from the CSV and successfully mapped the remaining fields (Name to the standard S Name field).

Problem 3: Dashboard UI Confusion

- The Issue: The instructions referenced a + Component button, but it was not visible.

- The Solution: I discovered Salesforce had updated their UI. The button is now labeled + Add Widget. I selected "Chart or Table" to add my reports.

Problem 4: Agentforce Testing

- The Issue: When prompted with "Summary of any Record", the agent asked for a specific ID instead of giving a general summary.

- The Solution: This proved the AI was working correctly and actively querying the database. By providing a specific ticket ID (e.g., TKT-001), the agent successfully returned the record data.

Problem 5: Successful Import

Final Result: After fixing the picklist values and remapping the CSV, I ran the Data Import Wizard again. The Bulk Data Load Jobs screen displayed Records Processed: 20 and Records Failed: 0. The data was successfully migrated.

6. ADVANTAGES & FUTURE SCOPE

Advantages

- Reduced Manual Effort: Automation handles the routing and prioritization instantly.
- AI-Ready: The system is integrated with Agentforce, demonstrating next-generation CRM capabilities.
- Data Integrity: Restricted picklists prevent bad data from entering the system.

Future Scope

- Email Alerts: Implement email notifications to the assigned agent when a High-priority ticket is created.
- Production Deployment: Migrate the configuration from the Developer sandbox to a live Production environment using Change Sets or Salesforce DX.
- Advanced AI: Train the Agentforce agent to automatically resolve basic "Billing" or "General" inquiries without human intervention.

7. CONCLUSION

The Customer Support Ticket Priority Prediction and Automation System successfully demonstrates how Salesforce can transform a manual, stressful process into an automated, data-driven workflow. By combining custom objects, robust flow automation, visual dashboards, and cutting-edge AI with Agentforce, the system ensures that critical support issues are never ignored and support teams can operate at maximum efficiency.

THANK YOU